

Module 3 DataBase Integration

Hands on -5: MongoDB

Task 1:

localhost:27017 > corporate_nosql Open MongoDB shell Create collection Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
feedback	-	36.86 kB	3.02 kB	11	274.00 B	1	36.86 kB

```
> db.feedback.insertMany([
  { "employee_id": 1, "course_code": "CS101", "rating": 5, "comments": "Excellent developer onboarding program.", "tags": ["clear", "engaging"], "semester": "2021-ODD", "status": "Completed" },
  { "employee_id": 2, "course_code": "CS101", "rating": 2, "comments": "Tech stack training was too fast-paced.", "tags": ["challenging", "fast-paced"], "semester": "2021-EVEN", "status": "Completed" },
  { "employee_id": 3, "course_code": "CS101", "rating": 4, "comments": "Good structural overview of project repos.", "tags": ["informative"], "semester": "2021-ODD", "status": "Completed" },
  { "employee_id": 4, "course_code": "CS102", "rating": 5, "comments": "Loved the microservices development module.", "tags": ["practical", "organized"], "semester": "2021-EVEN", "status": "Completed" },
  { "employee_id": 5, "course_code": "CS102", "rating": 3, "comments": "Decent framework overview, but needed deep-dives.", "tags": ["challenging"], "semester": "2021-ODD", "status": "Completed" },
  { "employee_id": 6, "course_code": "EC201", "rating": 1, "comments": "Documentation is dry and difficult to parse.", "tags": ["dry", "difficult"], "semester": "2021-EVEN", "status": "Completed" },
  { "employee_id": 7, "course_code": "EC201", "rating": 4, "comments": "Helpful code walkthroughs from senior staff.", "tags": ["helpful"], "semester": "2021-ODD", "status": "Completed" },
  { "employee_id": 8, "course_code": "MA301", "rating": 5, "comments": "Great guidance on engineering best practices.", "tags": ["excellent"], "semester": "2021-EVEN", "status": "Completed" },
  { "employee_id": 9, "course_code": "MA301", "rating": 2, "comments": "Hard criteria on pull request approvals.", "tags": ["tough"], "semester": "2022-ODD", "status": "In Progress" },
  { "employee_id": 10, "course_code": "CH101", "rating": 3, "comments": "Average resource planning session.", "tags": ["average"], "semester": "2021-EVEN", "status": "In Progress" }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a357383454ee04fe8a3bac4'),
    '1': ObjectId('6a357383454ee04fe8a3bac5'),
    '2': ObjectId('6a357383454ee04fe8a3bac6'),
    '3': ObjectId('6a357383454ee04fe8a3bac7'),
    '4': ObjectId('6a357383454ee04fe8a3bac8'),
    '5': ObjectId('6a357383454ee04fe8a3bac9'),
    '6': ObjectId('6a357383454ee04fe8a3baca'),
    '7': ObjectId('6a357383454ee04fe8a3bacb'),
    '8': ObjectId('6a357383454ee04fe8a3bacc'),
    '9': ObjectId('6a357383454ee04fe8a3bacd')
  }
}
```

With an attribute missing:

```
> db.feedback.insertOne({"employee_id":11,"course_code":"CS101","rating":4,"comments":"Useful starting point.","tags":["excellent","clear"],"semester":"2022-000"})
< {
  acknowledged: true,
  insertedId: ObjectId('6a3574c6454ee04fe8a3bad8')
}

> db.feedback.countDocuments
< [Function: countDocuments] AsyncFunction {
  apiVersions: [ 1, Infinity ],
  serverVersions: [ '4.0.3', '999.999.999' ],
  returnsPromise: true,
  topologies: [ 'ReplSet', 'Standalone', 'Sharded', 'LoadBalanced' ],
  returnType: { type: 'unknown', attributes: {} },
  deprecated: false,
  platforms: [ 'Compass', 'Browser', 'CLI' ],
  isDirectShellCommand: false,
  acceptsRawInput: false,
  shellCommandCompleter: undefined,
  newShellCommandCompleter: undefined,
  help: [Function (anonymous)] Help
}
```

Task 2:

```
> _MONGOSH
> db.feedback.find({rating:5})
< [
  {
    _id: ObjectId('6a3573a7454ee04fe8a3bace'),
    employee_id: 1,
    course_code: 'CS101',
    rating: 5,
    comments: 'Excellent developer onboarding program.',
    tags: [
      'clear',
      'engaging'
    ],
    semester: '2022-000',
    submitted_at: 2022-11-30T10:15:00.000Z,
    attachments: [
      {
        filename: 'onboarding_notes.pdf',
        size_kb: 240
      }
    ]
  },
  {
    _id: ObjectId('6a3573a7454ee04fe8a3bad1'),
    employee_id: 4,
    course_code: 'CS102',
    rating: 5,
    comments: 'Loved the microservices development module.',
    tags: [
      'practical',
      'organized'
    ]
  }
]
```

```

semester: '2022-ODD',
submitted_at: 2022-11-28T11:00:00.000Z,
attachments: [
  {
    filename: 'api_spec.zip',
    size_kb: 2048
  }
]
}
}
{
  _id: ObjectId('6a3573a7454ee04fe8a3bad5'),
  employee_id: 8,
  course_code: 'MA301',
  rating: 5,
  comments: 'Great guidance on engineering best practices.',
  tags: [
    'excellent'
  ],
  semester: '2022-ODD',
  submitted_at: 2022-11-25T10:00:00.000Z,
  attachments: [
    {
      filename: 'clean_code_rules.pdf',
      size_kb: 180
    }
  ]
}
}

```

```

> db.feedback.find({course_code:"CS101",tags:"challenging"})
< {
  _id: ObjectId('6a3573a7454ee04fe8a3bacf'),
  employee_id: 2,
  course_code: 'CS101',
  rating: 2,
  comments: 'Tech stack training was too fast-paced.',
  tags: [
    'challenging',
    'fast-paced'
  ],
  semester: '2022-ODD',
  submitted_at: 2022-12-01T09:00:00.000Z,
  attachments: [
    {
      filename: 'architecture_review.zip',
      size_kb: 1024
    }
  ]
}
> db.feedback.find({}, { employee_id: 1, course_code: 1, rating: 1, _id: 0 });
< {
  employee_id: 1,
  course_code: 'CS101',
  rating: 5
}
{
  employee_id: 2,
  course_code: 'CS101',

```

```
    employee_id: 6,  
    course_code: 'EC201',  
    rating: 1  
  }  
  {  
    employee_id: 7,  
    course_code: 'EC201',  
    rating: 4  
  }  
  {  
    employee_id: 8,  
    course_code: 'MA301',  
    rating: 5  
  }  
  {  
    employee_id: 9,  
    course_code: 'MA301',  
    rating: 2  
  }  
  {  
    employee_id: 10,  
    course_code: 'CH101',  
    rating: 3  
  }  
  {  
    employee_id: 11,  
    course_code: 'CS101',  
    rating: 4  
  }  
}
```

```
> db.feedback.updateMany({rating:{$lt:3}},{$set:{needs_review:true}})  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 3,  
  modifiedCount: 3,  
  upsertedCount: 0  
}
```

```
> db.feedback.updateMany({needs_review:true},{push:{tags:"reviewed"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

```
> db.feedback.deleteMany({semester:"2021-EVEN"})
< {
  acknowledged: true,
  deletedCount: 3
}
```

```
> db.feedback.aggregate([
  {$match:{semester:"2022-ODD"}},
  {$group:{_id:"$course_code", avg_rating:{$avg:"$rating"},total:{$sum:1}}},
  {$project :{ course_code:"$_id",_id:0,average_rating:{$round:["$avg_rating",1]},total:1}},
  {$sort: {avg_rating:-1}}
])
< {
  total: 2,
  course_code: 'CS102',
  average_rating: 4
}
{
  total: 2,
  course_code: 'CS101',
  average_rating: 3.5
}
{
  total: 2,
  course_code: 'MA301',
  average_rating: 3.5
}
{
  total: 1,
  course_code: 'EC201',
  average_rating: 1
}
```

TASK 3:

```
> db.feedback.aggregate([
  {$unwind:"$tags"},
  {$group:{_id:"$tags",count:{$sum:1}}},
  {$sort:{count:-1}}
])
< {
  _id: 'reviewed',
  count: 6
}
{
  _id: 'challenging',
  count: 2
}
{
  _id: 'excellent',
  count: 2
}
{
  _id: 'clear',
  count: 2
}
{
  _id: 'fast-paced',
  count: 1
}
{
  _id: 'engaging',
  count: 1
}
}
```

```
{
  _id: 'tough',
  count: 1
}
{
  _id: 'organized',
  count: 1
}
{
  _id: 'dry',
  count: 1
}
}
```

```
> db.feedback.createIndex({course_code:1})
< course_code_1
> db.feedback.find({course_code:"CS101"}).explain('executionStats')
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'corporate_nosql.feedback',
    indexFilterSet: false,
    parsedQuery: {
      course_code: {
        '$eq': 'CS101'
      }
    },
  },
  queryHash: '7AE81917',
  planCacheKey: '8C99514A',
  optimizationTimeMillis: 14,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  winningPlan: {
    stage: 'FETCH',
    inputStage: {
      stage: 'IXSCAN',
```

IXSCAN is confirmed instead of COLLSCAN